Dapagliflozin versus placebo in patients with heart failure and reduced ejection fraction: the CARDINAL-HF randomized trial

Halvorsen M, Okafor T, Lindqvist A, Ramires PL, Chen W. Cardiovascular Therapeutics Reports. 2021. doi:10.5555/ctr.2021.0101

Abstract

Background: Sodium-glucose cotransporter 2 (SGLT2) inhibitors may reduce heart failure events independent of glucose lowering. We tested dapagliflozin in patients with heart failure and a reduced ejection fraction.

Methods: In this double-blind, placebo-controlled trial, 4812 adults with New York Heart Association class II to IV heart failure and an ejection fraction of 40% or less were randomly assigned to dapagliflozin 10 mg once daily or placebo, in addition to guideline-directed therapy. The primary outcome was a composite of cardiovascular death or hospitalization for heart failure.

Results: Over a median of 18.4 months, the primary outcome occurred in 391 of 2410 patients (16.2%) in the dapagliflozin group and in 509 of 2402 patients (21.2%) in the placebo group (hazard ratio, 0.75; 95% CI, 0.66 to 0.85; P<0.001).

Conclusions: Among patients with heart failure and a reduced ejection fraction, dapagliflozin reduced the risk of cardiovascular death or hospitalization for heart failure compared with placebo.

Methods

Randomization was performed with a central interactive web response system, stratified by diabetes status, and allocation was concealed from investigators. Patients, investigators, and outcome assessors were unaware of the assigned treatment; placebo tablets matched dapagliflozin in appearance.

All deaths and hospitalizations were adjudicated by an independent clinical events committee unaware of treatment assignment. The prespecified statistical analysis plan was finalized before unblinding, and all outcomes were analyzed by intention to treat.

Results

Vital status was known for 99.8% of patients at the end of the trial. Baseline characteristics were balanced between groups; 42% of patients had type 2 diabetes.

Hospitalization for heart failure occurred in 236 of 2410 patients (9.8%) in the dapagliflozin group and in 322 of 2402 patients (13.4%) in the placebo group (hazard ratio, 0.71; 95% CI, 0.60 to 0.84).

Death from cardiovascular causes occurred in 232 of 2410 patients (9.6%) in the dapagliflozin group and in 276 of 2402 patients (11.5%) in the placebo group (hazard ratio, 0.83; 95% CI, 0.70 to 0.99).

The effect on the primary outcome was consistent in patients with and without diabetes (P for interaction=0.74).

Table 2. Outcomes

Outcome	Dapagliflozin	Placebo	HR (95% CI)
CV death or HF hospitalization	391/2410	509/2402	0.75 (0.66-0.85)
Hospitalization for HF	236/2410	322/2402	0.71 (0.60-0.84)
Cardiovascular death	232/2410	276/2402	0.83 (0.70-0.99)
Volume depletion (adverse event)	181/2410	163/2402	-

Safety

Adverse events related to volume depletion occurred in 7.5% of the dapagliflozin group and 6.8% of the placebo group. Major hypoglycemia and diabetic ketoacidosis were rare in both groups.

Discussion

Dapagliflozin added to guideline-directed therapy reduced cardiovascular death or hospitalization for heart failure by 25% in patients with a reduced ejection fraction. The number needed to treat to prevent one primary outcome event over 18 months was 20.

Synthetic document for offline testing. All names, identifiers, and numbers are invented.